

Reasoning Limitations of Multimodal Large Language Models. A Case Study of Bongard Problems

Mikołaj Małkiński¹ Szymon Pawlonka¹ Jacek Mańdziuk^{1,2}

Abstract

Abstract visual reasoning (AVR) involves discovering shared concepts across images through analogy, akin to solving IQ test problems. Bongard Problems (BPs) remain a key challenge in AVR, requiring both visual reasoning and verbal description. We investigate whether multimodal large language models (MLLMs) can solve BPs by formulating a set of diverse MLLM-suited solution strategies and testing 4 proprietary and 4 open-access models on 3 BP datasets featuring synthetic (classic BPs) and real-world (Bongard HOI and Bongard-OpenWorld) images. Despite some successes on real-world datasets, MLLMs struggle with synthetic BPs. To explore this gap, we introduce Bongard-RWR, a dataset representing synthetic BP concepts using real-world images. Our findings suggest that weak MLLM performance on classical BPs is not due to the domain specificity, but rather comes from their general AVR limitations. Code and dataset are available at: <https://github.com/pavonism/bongard-rwr>

1. Introduction

Analogy-making is a critical aspect of human cognition, tightly linked with fluid intelligence, the capacity to apply learned skills in novel settings (Lake et al., 2017). Several approaches have been proposed to build systems capable of making analogies. Notably, the structure-mapping theory explores methods for discovering structural correspondences between pre-existing object representations (Winston, 1982; Gentner, 1983; Carbonell, 1983; Falkenhainer et al., 1989; Holyoak & Thagard, 1989). However, these approaches often overlook the perceptual aspect, assuming object rep-

¹Warsaw University of Technology, Warsaw, Poland ²AGH University of Krakow, Krakow, Poland. Correspondence to: Mikołaj Małkiński <mikolaj.malkinski.dokt@pw.edu.pl>.

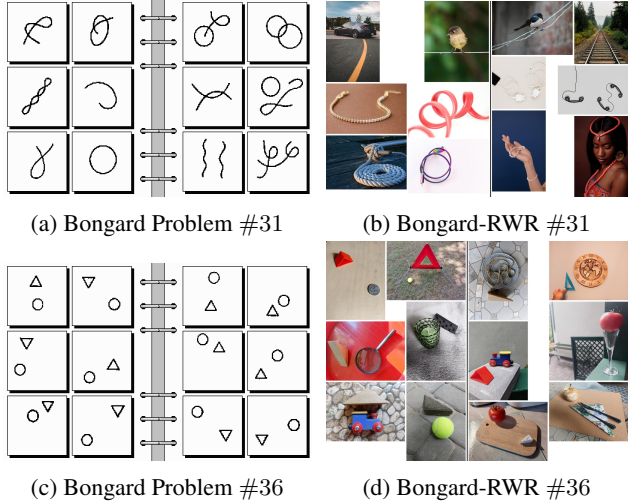


Figure 1: **Bongard Problems.** (a), (c): Manually-designed synthetic BPs (Bongard, 1970). (b), (d): Their real-world equivalents from the Bongard-RWR dataset proposed in this work. BP #31: Left: One line. Right: Two lines. BP #36: Left: Triangle above circle. Right: Circle above triangle.

resentations are already given. Chalmers et al. (1992) highlight that forming useful representations is an intricate challenge. In particular, perception is not merely a passive reception of sensory data, but rather an active interpretation influenced by prior knowledge. This process involves the detection of patterns, recognition of analogies, and abstraction of concepts. The resultant representations may vary significantly depending on the context, which underscores the importance of modeling perception and cognition jointly (Hofstadter, 1995).

Multiple problems that necessitate combined perception and reasoning have been identified (Hofstadter, 1999). Among these tasks are Bongard Problems (BPs), introduced by Bongard (1968; 1970). Initial BPs were designed manually, leading to the formulation of a few hundred task instances by individual contributors (Foundalis, 2006b). A typical BP consists of two sides, left and right, each comprising six image panels arranged in a grid (Figs. 1a, 1c). All images on one side illustrate a shared concept absent in the

images on the opposite side. The task is to identify the underlying rule that differentiates the sides and **articulate it in natural language**. Initial BPs (Bongard, 1968), akin to human IQ tests, featured abstract 2D geometric shapes, putting the focus on abstract reasoning. However, recent works have expanded the set of BPs to include real-world images, which broadens the scope of presented objects, attributes and relations. Specifically, the matrices in Bongard HOI (Jiang et al., 2022) depict human-object interactions, while Bongard-OpenWorld (Wu et al., 2024) employs open-world free-form concepts, increasing the diversity of featured scenes.

A central theme in BPs is recognition of concepts in a context-dependent manner, as object representations need to be formed specifically for the presented matrix, rather than described *a priori* (Linhares, 2000). For example, consider the matrix in Fig. 1a – an analysis restricted to its left side may yield multiple concepts, such as the presence of curves or an object centered in the image. Only through a comprehensive understanding of both matrix sides one can recognize that the left side depicts a single line, while the right side presents two lines. Such concept-based tasks are argued to offer a more accurate evaluation of the model’s generalization ability and its capacity for abstraction (Mitchell, 2021; Odouard & Mitchell, 2022). In contrast to other abstract reasoning problems, such as Raven’s Progressive Matrices (RPMs) (Raven, 1936; Raven & Court, 1998; Małkiński & Mańdziuk, 2025a) that have recently witnessed the development of large-scale benchmarks (Barrett et al., 2018; Zhang et al., 2019; Małkiński & Mańdziuk, 2025c), BPs allow to assess model’s ability to derive concepts from a limited set of examples (typically six images per matrix side), which positions the task within a few-shot learning setting (Fei-Fei et al., 2006; Wang et al., 2020). The above aspects make BPs a valuable testbed for assessing abstract reasoning abilities of AI models.

Motivation. The quest to build systems capable of forming abstract concepts dates back to the 1950s (McCarthy et al., 2006). The advent of Deep Learning (DL) opened new possibilities to tackle BPs (Kharagorgiev, 2018; Nie et al., 2020). However, despite significant advancements, methods for consistently solving BPs (and other problems that involve abstract reasoning) are still lacking (Mitchell, 2021; van der Maas et al., 2021; Stabinger et al., 2021). Typically, DL approaches omit the generation of natural language answers by casting BP into a binary classification task, in which a test image had to be assigned to the matching side of the matrix. Conversely, a parallel stream of research on large language models (LLMs) demonstrated promising results in open-ended language generation (Brown et al., 2020). In particular, LLMs were applied to selected AVR tasks (Webb et al., 2023), though, lately Xu et al. (2024)

pointed certain LLM limitations in solving AVR problems represented as text despite using information lossless translation through direct-grid encoding. Recent works have combined the vision and language modalities into multimodal large language models (MLLMs) (Achiam et al., 2023; Reid et al., 2024; Anthropic, 2024), inviting their application to diverse tasks (Yin et al., 2024; Wu et al., 2023). Motivated by these recent developments we examine the reasoning capabilities of MLLMs in solving BPs.

Contribution. The contribution of this paper is fourfold.

- (1) We consider BPs in the context of MLLMs and propose a diverse set of strategies to solve BP instances in two setups: open-ended language generation and binary classification.
- (2) We evaluate 4 state-of-the-art proprietary MLLMs and 4 open MLLMs on both synthetic and real-world BPs, and identify their severe abstract reasoning limitations.
- (3) To further examine the observed MLLM limitations in solving both synthetic and real-world BPs, we introduce a focused dataset of BPs (Bongard-RWR, Figs. 1b, 1d) that represent concepts from synthetic BPs using real world images. Thanks to relying on **the same abstract concepts** as synthetic BPs, Bongard-RWR facilitates direct comparisons of the MLLM performance in both domains.
- (4) We perform a detailed comparative analysis of 8 MLLMs on Bongard-RWR vs. synthetic BPs, shedding light on the reasons of their generally poor performance.

2. Related Work

AVR Tasks. The AVR field encompasses a broad set of problems aimed at studying various aspects of visual cognition (Gardner & Richards, 2006; Małkiński & Mańdziuk, 2023). Recent DL research in this domain gravitated towards utilizing certain well-established datasets, e.g. with visual analogies (Hill et al., 2019; Webb et al., 2020) or RPMs (Zhang et al., 2019; Barrett et al., 2018), to measure the progress of DL models (Małkiński & Mańdziuk, 2024a;b; 2025b). However, such benchmarks evaluate system performance in learning a particular task, rather than assessing its general ability to acquire new AVR skills. To address this limitation, certain tasks have adopted few-shot learning setups, requiring models to learn from a few demonstrations, as exemplified by SVRT (Fleuret et al., 2011) or Bongard-LOGO (Nie et al., 2020). Nonetheless, these benchmarks follow a discriminative setting where a set of possible answers is provided. Conversely, other datasets such as ARC (Chollet, 2019) or PQA (Qi et al., 2021) pose a generative challenge, which may be considered more difficult due to its open-ended nature. In addition to synthetic tasks featuring 2D geometric shapes, certain datasets present analogous reasoning tasks using real-world images (Teney

et al., 2020; Ichien et al., 2021; Bitton et al., 2023). This approach extends the range of concepts that can be expressed and, above all, allows employing models pre-trained on large image datasets. In this work, we concentrate on several BP datasets that present a few-shot learning challenge, cover both synthetic and real-world images, and consider settings involving both binary classification and answer generation in natural language.

Approaches to solve BPs. Initial approaches to tackle BPs involved cognitive architectures (Foundalis, 2006a), program synthesis coupled with inductive logic programming (Saito & Nakano, 1996; Sonwane et al., 2021), and the application of Bayesian inference within a visual language framework (Depeweg et al., 2018; 2024). (Kharagorgiev, 2018) trained a convolutional network on a generated synthetic dataset with geometric shapes and applied a one-level decision tree to solve BPs framed as a binary classification task. Nie et al. (2020) introduced Bongard-LOGO with synthetically generated BPs and used it to evaluate CNN-based models focused on meta-learning (Snell et al., 2017; Mishra et al., 2018; Lee et al., 2019; Raghu et al., 2020; Chen et al., 2021) and relational reasoning (Barrett et al., 2018). Jiang et al. (2022) applied the Relation Network (Santoro et al., 2017) to objects detected with Faster R-CNN (Ren et al., 2015) and employed the model to solve matrices from the real world Bongard HOI dataset. Despite high diversity of approaches, none of them has fully addressed the abstract and open-ended nature of BPs. Most related to our work, Wu et al. (2024) considered hybrid approaches that caption each image panel with an image-to-text model and applied LLMs for processing these text descriptions. Differently, in this work we focus on MLLMs that are inherently capable of jointly processing images and text.

Abstract reasoning of MLLMs. While the abstract reasoning capabilities of MLLMs have only recently begun to be explored, they have been studied more extensively in several related tasks. Initial works focused on LLMs and evaluating their abstract reasoning performance in simplified analogy tasks. Webb et al. (2023) showed that GPT-3 and GPT-4 (text-only variants) performed on the human level, or even outcompeted humans, in certain RPM-like tasks in a zero-shot manner without additional fine-tuning. However, they represented the image objects as text using a fixed small vocabulary, thus omitting the need for identifying concepts from open-ended shapes, a key challenge of BPs. Recent research concerning the evaluation of abstract reasoning skills of LLMs concentrates around the Abstraction and Reasoning Corpus (ARC) task (Chollet, 2019). It was demonstrated that LLMs can solve certain ARC problems transformed to the text domain (Moskvichev et al., 2023; Mirchandani et al., 2023; Camposampiero et al., 2023; Xu et al., 2024). Despite these important stepping stones, the text-based rep-

resentation taken in these works simplifies the perception task by presenting the model with pre-existing higher level representations. Only recently, thanks to the appearance of MLLMs, vision and text started to be treated jointly in a unified manner. Perception Test (Patraucean et al., 2023) assessed multimodal video understanding across various skills and types of reasoning. Cao et al. (2024) proposed a suite of AVR tasks to compare MLLM and human performance. Jiang et al. (2024) assessed AVR skills of MLLMs on an introduced multidimensional benchmark combining AVR and perceptual questions. KiVA (Yiu et al., 2025) evaluated how models generalize visual transformations to analogical reasoning tasks inspired by the developmental psychology. Most relevant to our paper is the concurrent work of Wüst et al. (2024), which investigated the capabilities of MLLMs in solving synthetic BPs, using both open-ended text generation and multi-class concept classification setups. Our work complements this stream of research by exploring BPs and providing comparative insights into MLLM analogy-making performance in both synthetic and real-world domains.

3. Solving BPs with MLLMs

In this paper, we propose a set of novel strategies for solving BPs using MLLMs. Definition of each strategy includes the input on which the model operates and the sequence of reasoning steps performed by the model. A high-level overview of these methods is provided in Fig. 2. In the main tested setting, we follow the initial BP formulation that requires providing **an answer in natural language**, and propose a model-based approach to automatically evaluate such model predictions. In addition, we consider simpler formulations of the problem, casting it into a binary classification framework that enables detailed evaluation of AVR abilities of the tested MLLMs. An illustration of these evaluation settings is presented in Fig. 3. In what follows, let $\mathcal{BP}^X = \{\mathcal{L}^X, \mathcal{R}^X, y^X\}$ denote a BP instance ($X \in \mathcal{N}$ is an index), composed of $\mathcal{L}^X = \{L_1^X, \dots, L_6^X\}$ left and $\mathcal{R}^X = \{R_1^X, \dots, R_6^X\}$ right panels, resp., and its concept y^X expressed in natural language.

3.1. Answer Generation in Natural Language

We start by defining the strategies for generating answers in natural language. In each strategy, the model receives a general description of Bongard Problem with two BP examples with correct answers. Additionally, besides this generic introductory information, a given task $\mathcal{BP}^X$ to be solved is presented in a **strategy-specific** way. Appendix I.4 presents the exact prompt formulations.

Direct (Fig. 2a). The model receives an image presenting $\mathcal{BP}^X$ and is asked to directly formulate an answer (i.e., describe the difference between $\mathcal{L}^X$ and $\mathcal{R}^X$ panels in natural language).